Instrumentation and sensors for the food industry

Second edition

Edited by

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Symbols

Chapter 13

c	speed of sound
υ	dispersive ultrasound phase velocity
Z	specific acoustic impedance
α	coefficient of ultrasound attenuation
α_{T}	coefficient of transmission through boundary
f	frequency
λ	wavelength
ω	angular frequency; $\omega = 2\pi f$
k	wave vector; $k = 2\pi/\lambda$
τ	relaxation time
ρ	density
P	absolute pressure
M	molecular weight
η	coefficient of viscosity
r_{p}	radius of suspended spherical particle
V	volume
T	absolute temperature
E_{M}	elastic modulus (as appropriate)
K	bulk modulus
Y	Young's modulus
G	shear modulus
K_{S}	adiabatic bulk modulus
β_P	coefficient of expansion at constan pressure
γ	ratio of principal specific heats c_P/c_V
c_P/c_V	specific heats at constant pressure, volume
κ	coefficient of thermal conductivity
R	universal gas constant
B	second virial coefficients
STP	standard temperature and pressure

Chapter 15

A_n	function in equation (15.22)
B_n	function in equation (15.22)
d	capillary tube diameter
e	distance from bottom of bob to cup or container
f	frequency
G	shear modulus
G'	storage modulus
G''	loss modulus
g	gravitational constant
H	separation of the two plates
h	difference in levels between liquid in reservoirs of a capillary viscometer
h_c	end effects correction in concentric cylinder viscometry
h'	height of bob for a rotary viscometer
I_2	Bessel function
K	constant for capillary viscometer
K_n	root of modifier Navier-Stokes equation
k	apparent viscosity or consistency index for power law fluid
k'	constant in Casson equation
k''	constant in Herschel-Buckley
L	length of capillary tube
L_o	sample length before deformation
L'_o	final sample length after deformation
n	power law exponent
Δp	pressure drop along a capillary tube
Q	flow rate
R_c	radius of cone
R_1	radius of bob in concentric cylinder viscometer
R_2	radius of cup in concentric cylinder viscometer
R_p	radius of the plate (parallel plate systems)
ΔR	$R_2 - R_1$
T	torque
T_e	equivalent torque
t	time (equations (15.8) (15.9) or (15.12))
t'	relaxation time (equation (15.4))
u_p	slope of plot of shear stress versus shear rate once yield stress has been exceeded
V	volume of liquid used in a capillary viscometer
α	conical angle
β	relaxation time
γ	shear strain $(= (L'_o - L_o)/L_o$, dimensionless)
$\dot{\gamma}$	shear rate (or rate of shear strain) $\mathrm{d}\gamma/\mathrm{dt}$
$\dot{\gamma}_{eff}$	effective shear rate
$\dot{\gamma}_{max}$	shear rate at the outer edge (rim) of the plate
$\dot{\gamma}_{meas}$	measured shear rate
δ	phase displacement angle
μ	viscosity
μ_0	viscosity at zero shear rate

μ_p	slope of straight line plot of shear stress vs. shear rate (equation 15.17)
μ_∞	viscosity at infinite shear rate
ω	angular velocity
π	3.1417
ρ	density
τ	shear stress
τ_b	shear stress at the bob
τ_y	yield stress

Chapter 17

b	number of bits
c	mass concentration of medium, g ml^{-1}
f	frequency, Hz
k	initial specific conductivity, S m^{-1}
l	effective separation of electrodes, m
n	population density, cfu ml^{-1}
n_0	inoculum density, cfu ml^{-1}
n_s	maximum possible population density, cfu ml^{-1}
n_D	arbitrary detection level of population density, cfu ml^{-1}
$\dot{n}$	growth rate, cfu h^{-1}
$\dot{n}/n$	specific growth rate, h^{-1}
$(\dot{n}/n)_0$	initial specific growth rate, h^{-1}
r	correlation coefficient
t	time, h
t_g	generation time, h
t_g'	$1/(\dot{n}/n)_o$, h
t_D	time after inoculation at which population density grows to a value n_D, detection time, h
t_L	lag time, h
A	electrode area, m^2
A_D	conversion gain
C_D, R_D	series capacitance and series resistance due to alignment of polar dipoles in double charge layer in fluid, F, Ω
C_{ox}, R_{ox}	series capacitance and series resistance resulting from the presence of an oxide layer at the electrode surfaces, F, Ω
C_{se}	lumped electrode capacitances C_D and C_{ox} (for R_D, $R_{ox} \ll R_s$), F
G_0	initial conductance of medium in test cell, S
G_s	subsequent conductance of medium in test cell, S
K	increase in specific conductance associated with production of single bacterial cell, per unit volume, S m^{-1} (cfu ml^{-1}) ^{1}
N	number of colonies counted on agar plate
R_{fs}	full-scale range of resistance, Ω
R_s	true resistance of medium in test cell, Ω
S	substrate mass, g
S_0, S_s	initial and final stationary mass of substrate, g
ΔS	mass of substrate metabolised by one bacterial cell, g

V, i	instantaneous voltage, current, V, A
$\|V\|$,$\|i\|$	peak voltage, current, V, A
$\|V_{\mathrm{m}}\|$,$\|i_{\mathrm{m}}\|$	mean voltage, current, modulus, V, A
X_{s}	reactance of C_{s}, Ω
X_{se}	reactance of C_{se}, Ω
Y	admittance, S
Z	impedance of R_{s} and C_{s} in series, Ω
ϕ	phase angle
γ_D	number of generations after inoculation at which growth is detected